

KNZiN Feature 005 Status Report

Learner Hub / Course Library / Ticket Ledger

Branch: 005-learner-hub

Date: October 01, 2026

Audience: Project Leadership & Technical Review

IMPLEMENTATION STATUS

READY

PRODUCTION STATUS

PREREQUISITES PENDING

EXECUTIVE STATUS

Feature 005 has completed implementation, repository/local-runtime verification, security verification, clean-database migration validation, and final reconciliation. The work remains uncommitted on the feature branch for human review. The code is ready for review and release preparation. Production launch is not yet cleared because the external operational prerequisites listed on Page 2 remain pending.

DELIVERED SCOPE

- **Learner Dashboard:** Curriculum & owned-scope progress at `/[locale]/dashboard`.

■ **Gated Part Access:** Free Part 1 preview and authenticated, entitlement-gated paid parts.

■ **Identifier Generation:** Server-generated Crockford Base32 learner codes & canonical serials.

■ **Ticket Ledger Drawer:** Accessible sliding drawer with bidirectional Arabic RTL & English LTR.

■ **Account Continuity:** Guest-to-Google merge preserving orders, entitlements, tickets, and progress.
- **Server Entitlements:** Authoritative single-part access and full-course bundles.

■ **Media Protection:** Private storage with short-lived 15-minute signed streaming & download access.

■ **Ticket Ledger:** Asynchronous fulfillment, idempotency, recovery, dynamic draw eligibility.

■ **Monotonic Progress:** Server-validated watch tracking with sticky 95% completion behavior.

■ **Fulfillment Simulator:** Local non-production simulation command for end-to-end verification.

VERIFICATION RESULTS

VERIFICATION AREA	VERIFIED OUTCOME & METRIC DETAILS	STATUS & BOUNDARY
Backend Automated Suite	85 passed, 0 failed, 5,739 assertions	PASS Full local run (34.57s)
Backend Test Inventory	22 files / 22 classes (20 Feature + 2 Unit)	VERIFIED Exact filesystem count
Frontend Automated Suite	63 passed, 0 failed, 21 test suites	PASS Jest DOM suites
Production Build	Compiled cleanly via <code>npm run build</code>	PASS 0 TypeScript errors
Static Pages Generated	47 static routes prerendered	VERIFIED Includes <code>/[locale]/dashboard</code>
Clean Database Migration	5 Feature 005 migrations run on isolated clean <code>knzin_test</code>	VERIFIED No scratch schema repair
Migration Rollback Integrity	Rollback of 5 steps and subsequent re-migration succeed cleanly	VERIFIED Clean drop & re-creation
Security Attack Scenarios	24 defined attack vectors verified (playback, downloads, tickets, progress, merge)	PASS Server-enforced boundaries
Concurrency Verification	Design/DB guarantees + sequential simulation (missing delta, duplicate worker)	NO PARALLEL TEST CLAIM
Integration Verification	Local runtime daemons (MariaDB 3306, API 8000, Web 3000) + curl HTTP 200	NO BROWSER-E2E CLAIM

SECURITY & RELIABILITY CONTROLS

- 1. Server-Authoritative Access:** Paid-resource access is authorized server-side through authenticated entitlement checks. Client-side purchase state in `localStorage` is untrusted.
- 2. Protected Media Origin:** Paid media is served strictly through private storage and short-lived signed URLs, not as public or unlisted YouTube content.
- 3. Signed URL Validity:** Playback and download URLs are time-limited bearer capabilities with a maximum validity of 15 minutes (900 seconds).
- 4. Ticket Ledger Idempotency:** Minting uses MariaDB sequence allocation, row-level locking (`lockForUpdate`), unique index enforcement (`uq_order_ticket_index`), and exact missing-delta calculations.
- 5. Monotonic Progress Persistence:** Progress reports are validated server-side using monotonic `GREATEST()` checks. Reaching 95% permanently locks completion status.
- 6. Deterministic Account Merge:** One-way guest-to-Google merge transfers entitlements, progress, and tickets under ascending row locks, linking duplicate active scopes as superseded.
- 7. Dynamic Watermarking:** Floating canvas displays learner email, code, and timestamp over video elements. Serves as psychological deterrence and forensic tracing, not absolute capture prevention.

PRODUCTION RELEASE PREREQUISITES (PHASE 9 GATES)

- 1. T058 (Storage Provisioning):** Provision private storage for paid video assets on configured protected-media provider (AWS S3 / Cloudflare R2 / Bunny Storage).
- 2. T059 (Origin Security):** Verify storage origin rejects anonymous public HTTP access (confirming HTTP 403 without valid signature).
- 3. T060 (YouTube Lockdown):** Set legacy Part 2+ YouTube assets to **Private or Deleted** in YouTube Studio. **Unlisted is strictly prohibited** as it does not enforce entitlement boundaries.
- 4. T061 (Database Migration):** Execute Feature 005 migrations (2026_10_01_000001 to 000005) in the production database environment.
- 5. T062 (Frontend & CDN Cutover):** Deploy production frontend build and execute edge CDN cache purge (Cloudflare / Vercel) to invalidate cached bundles containing obsolete logic.
- 6. Queue Worker Infrastructure:** Configure production Redis queue and run an active worker processing `App\Jobs\GenerateTicketsJob`.
- 7. Reconciliation Scheduler:** Execute `php artisan schedule:run` every minute via system cron; ticket reconciliation is configured to run **every 5 minutes**.

VERIFICATION BOUNDARIES & TECHNICAL SCOPE LIMITATIONS

KNOWN BOUNDARIES & NON-CLAIMS

- **Concurrency:** No true multi-process/parallel race test was executed under PHPUnit. Concurrency relies on MariaDB transaction isolation, row locking, and unique constraints, verified via sequential worker simulations.
- **Browser E2E:** No automated Playwright or Cypress browser-driven end-to-end test suite was executed.
- **Deployment Dependencies:** Production media storage, queue worker, scheduler cron, database migration, and edge CDN cache purge remain mandatory deployment gates.
- **Catalog Evolution Policy:** Automatic entitlement inclusion of newly published course parts under historical bundle purchases remains an open, unresolved product decision.

REVIEW & REPOSITORY STATE

Branch: 005-learner-hub

Working Tree: Clean, reviewed by Product Owner

Evidence Map: `specs/005-learner-hub/evidence/INDEX.md`

Test Plan: `specs/005-learner-hub/test-plan.md`

NEXT FEATURE PIPELINE (PLATFORM TRAJECTORY)

Completed & Verified: 001 UI Foundation · 002 Auth & Catalog · 003 Draws Arena · 004 Trust & Engagement · 005 Learner Hub
Next in Pipeline: **006 Multi-Tier Affiliate Engine** → 007 Payment Gateway → 008 Admin Operations → 009 Multi-Channel Notifications

FINAL SIGN-OFF STATUS

Feature 005 implementation is complete and verified at repository and local-runtime level.
Production launch remains pending until the 7 operational release prerequisites are completed and signed off by operations.